

**PRESENTED BY :**

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TEAM G-1

**OMNICHANNEL RETAIL  
SALES AND INVENTORY  
ANALYTICS PROJECT**

2026

2026

2026

2026

2026

2026

2026

**DATA ANALYTICS**

# TEAM G-1

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SMITA**



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ZAKY  
MUQODDAS**



# TEAM G-1

# DATA UNDERSTANDING

## Omnichannel Retail Sales and Inventory Dataset

This project analyzes an AI generated dataset of omnichannel retail business by integrating data across customers, orders, products, inventory, and warehouses.

## Data Scope

The dataset consist of 2023-2024 retail transactions (50K rows) across 51 branch store and 10 warehouses in India.

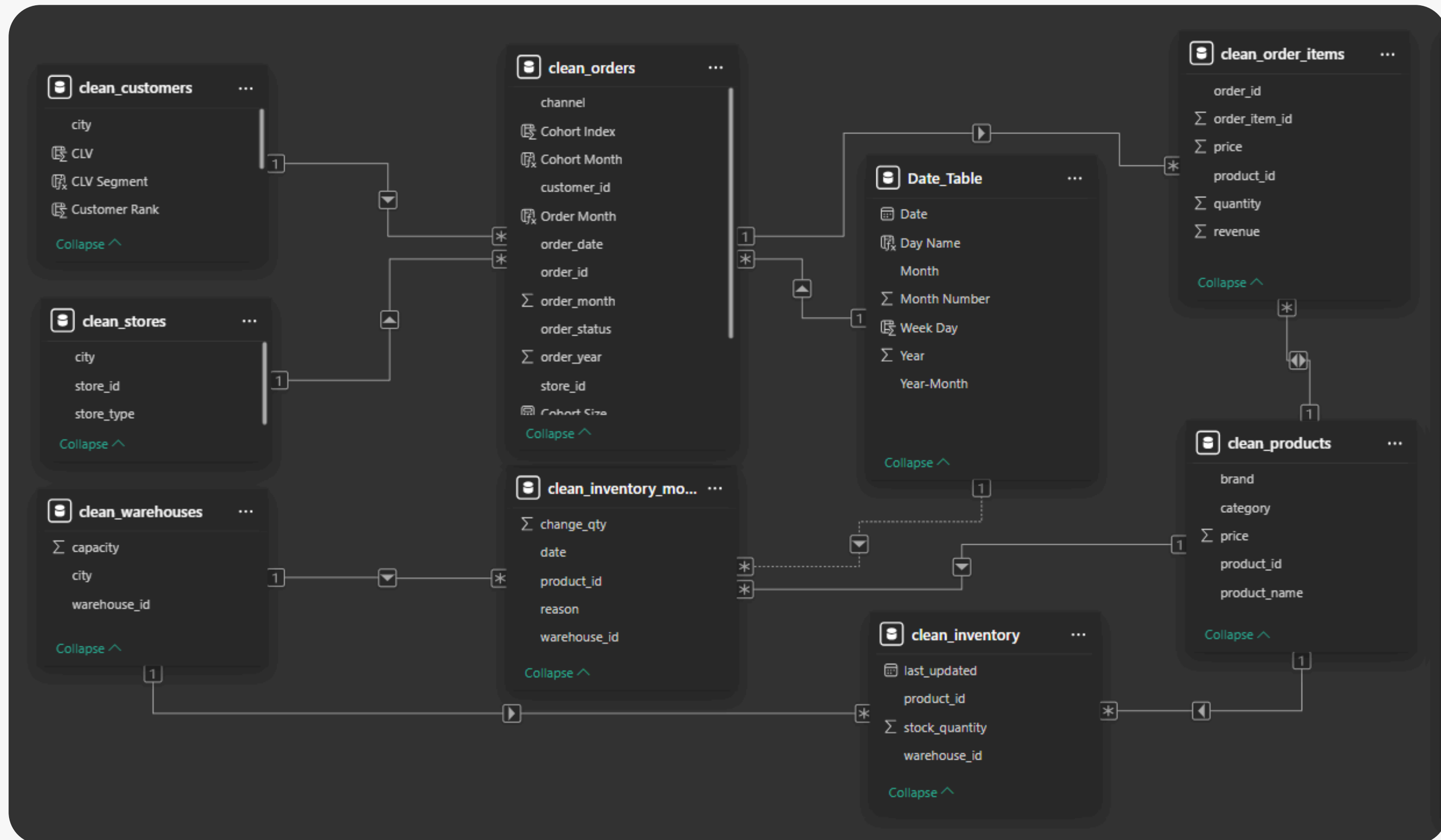
## Data Relation

The dataset consist of 8 related tables including : customers, orders, order\_items, products, inventory, inventory\_movements, stores, and warehouses

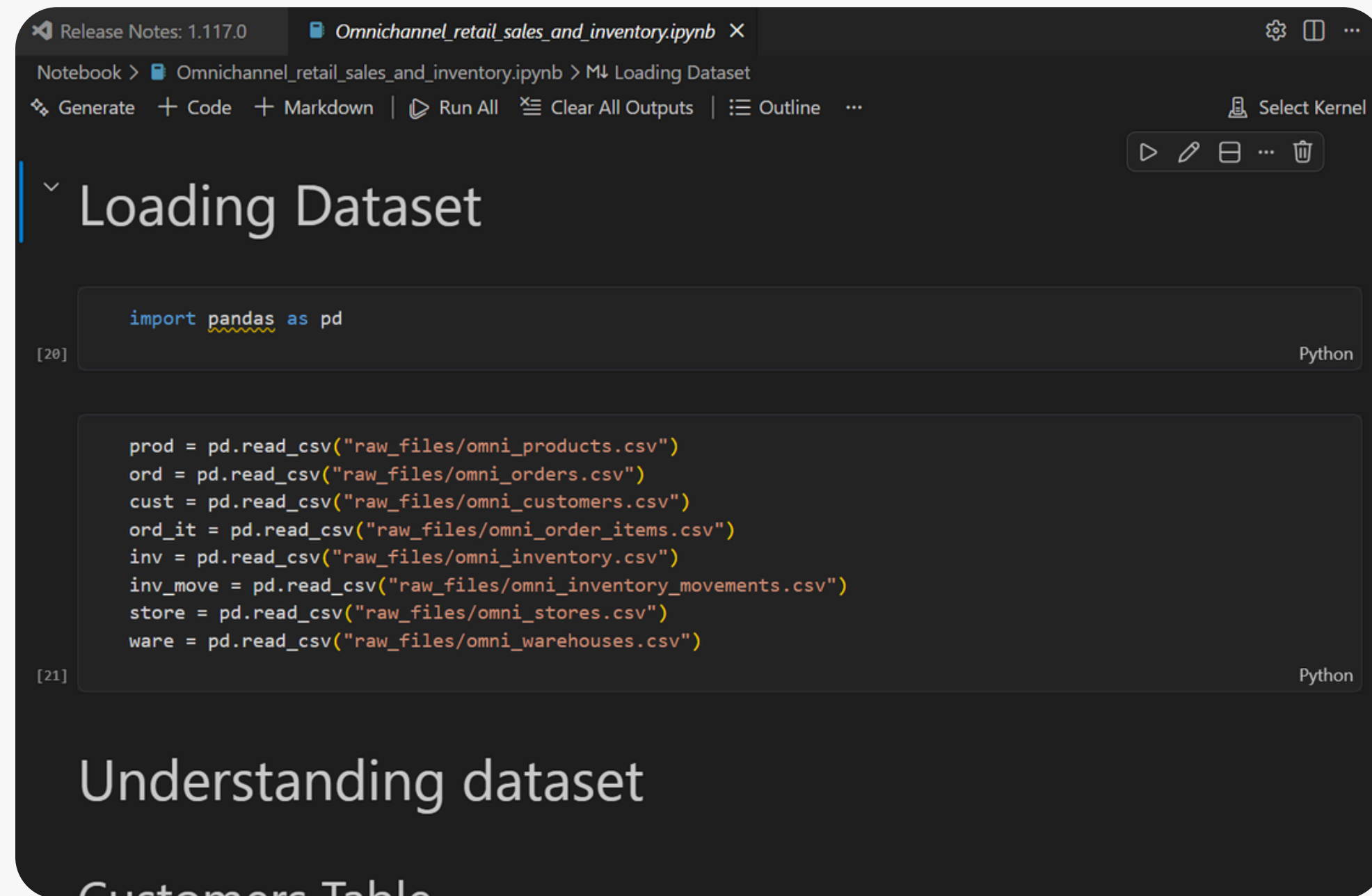
## Project Goal

- Understand sales performance across channels (Store, Online, App)
- Track inventory movements and stock levels
- Analyze customer behavior and retention
- Build a scalable data model for business intelligence

# DATA DIAGRAM



# PYTHON DATA CLEANING



The screenshot shows a Jupyter Notebook interface with the following elements:

- Tab:** Omnichannel\_retail\_sales\_and\_inventory.ipynb
- Header:** Notebook > Omnichannel\_retail\_sales\_and\_inventory.ipynb > M Loading Dataset
- Toolbar:** Generate, + Code, + Markdown, Run All, Clear All Outputs, Outline, Select Kernel.
- Section Header:** Loading Dataset
- Code Cell [20]:**

```
import pandas as pd
```
- Code Cell [21]:**

```
prod = pd.read_csv("raw_files/omni_products.csv")
ord = pd.read_csv("raw_files/omni_orders.csv")
cust = pd.read_csv("raw_files/omni_customers.csv")
ord_it = pd.read_csv("raw_files/omni_order_items.csv")
inv = pd.read_csv("raw_files/omni_inventory.csv")
inv_move = pd.read_csv("raw_files/omni_inventory_movements.csv")
store = pd.read_csv("raw_files/omni_stores.csv")
ware = pd.read_csv("raw_files/omni_warehouses.csv")
```
- Section Header:** Understanding dataset
- Text:** Customers Table

- Standardized column names and its data types
- Removed duplicates and invalid values
- Converted date formats
- Handled missing values (e.g. online store mapping)
- Fixed inventory movement logic (positive/negative quantities)
- Created derived columns (revenue, order\_year, order\_month)
- Primary and foreign key validation
- etc

PANDAS

NUMPY

MATPLOTLIB

SEABORN

SQLALCHEMY

PANDAS

NUMPY

MATPLOTLIB



# POSTGRESQL DATA ANALYSIS

Query

Query History

247

sum(freq\_2023) freq\_2023, sum(freq\_2024) freq\_2024

248

from customer\_segment

249

group by 1 order by 2 desc

250

/\*

251

COHORT ANALYSIS

252

Based on signup\_date and valid trans, number of membered customer decreased significantl

253

Meanwhile Legacy segment still loyal to company showed by increased transaction frequenc

254

opposite even though the difference very small.

255

\*/

256

257

258

-- 11. Sales to Stock Ratio

259

with monthly\_sales as (

Data Output

Messages

Notifications

≡+

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▼

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⬇

📈

SQL

Showing rows: 1 to 3

✎

Page No: 1

|   | segment<br>text | total<br>bigint | freq_2023<br>numeric | freq_2024<br>numeric |
|---|-----------------|-----------------|----------------------|----------------------|
| 1 | Legacy          | 2085            | 5232                 | 5256                 |
| 2 | New 2023        | 1975            | 4916                 | 4882                 |
| 3 | New 2024        | 915             | 2309                 | 2260                 |

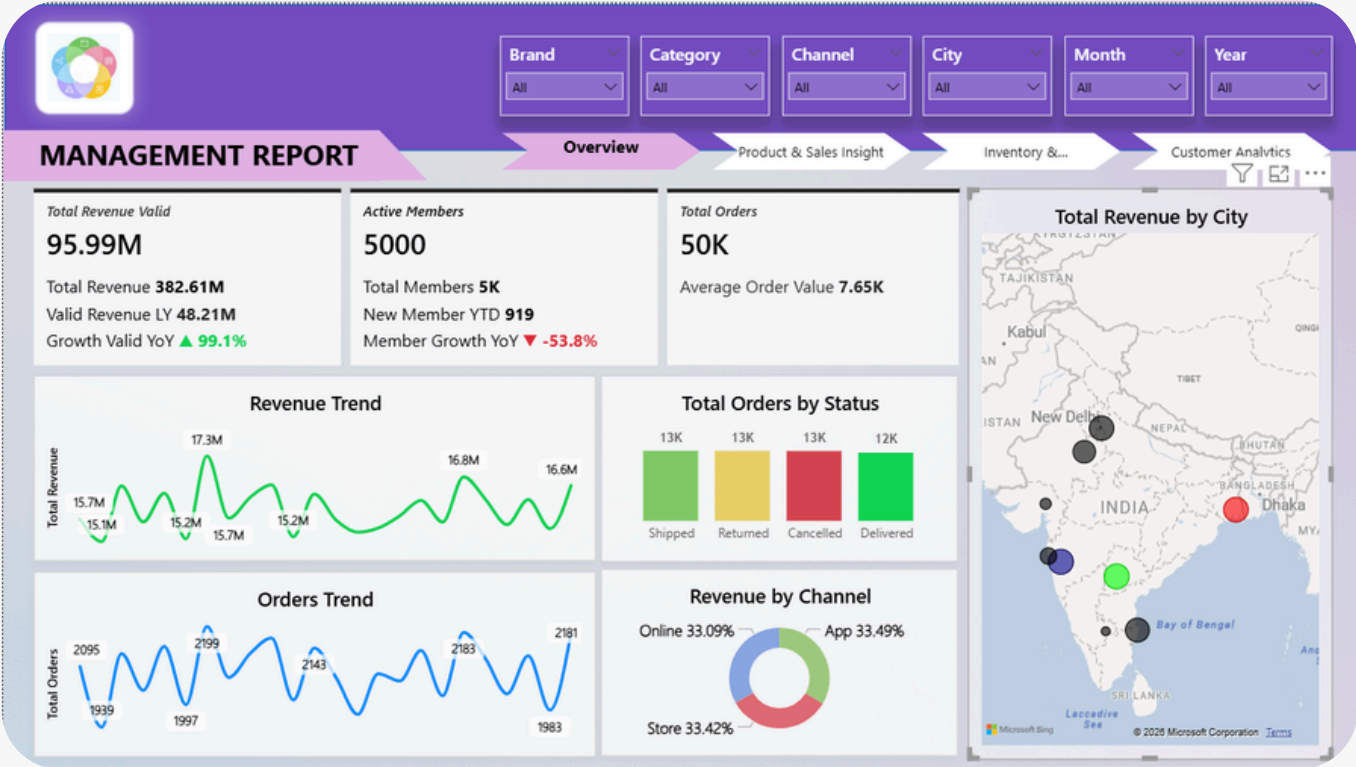
Total rows: 3

Query complete 00:00:00.093

- Revenue Over Time
- Best Sales Product over Time
- Cancelled and Returned Product over Time
- Product Performance by Revenue
- Store and City Performance by revenue
- Online Channel Comparison by Revenue
- Basket Analysis by Average Order Value
- Customer Loyalty by Order Count and Revenue Generated
- Customer Distribution and Its Revenue Generated
- Cohort Analysis
- Sales to Stock Ratio
- Inventory Turn Over Ratio

# POWER BI DASHBOARD

## OVERVIEW



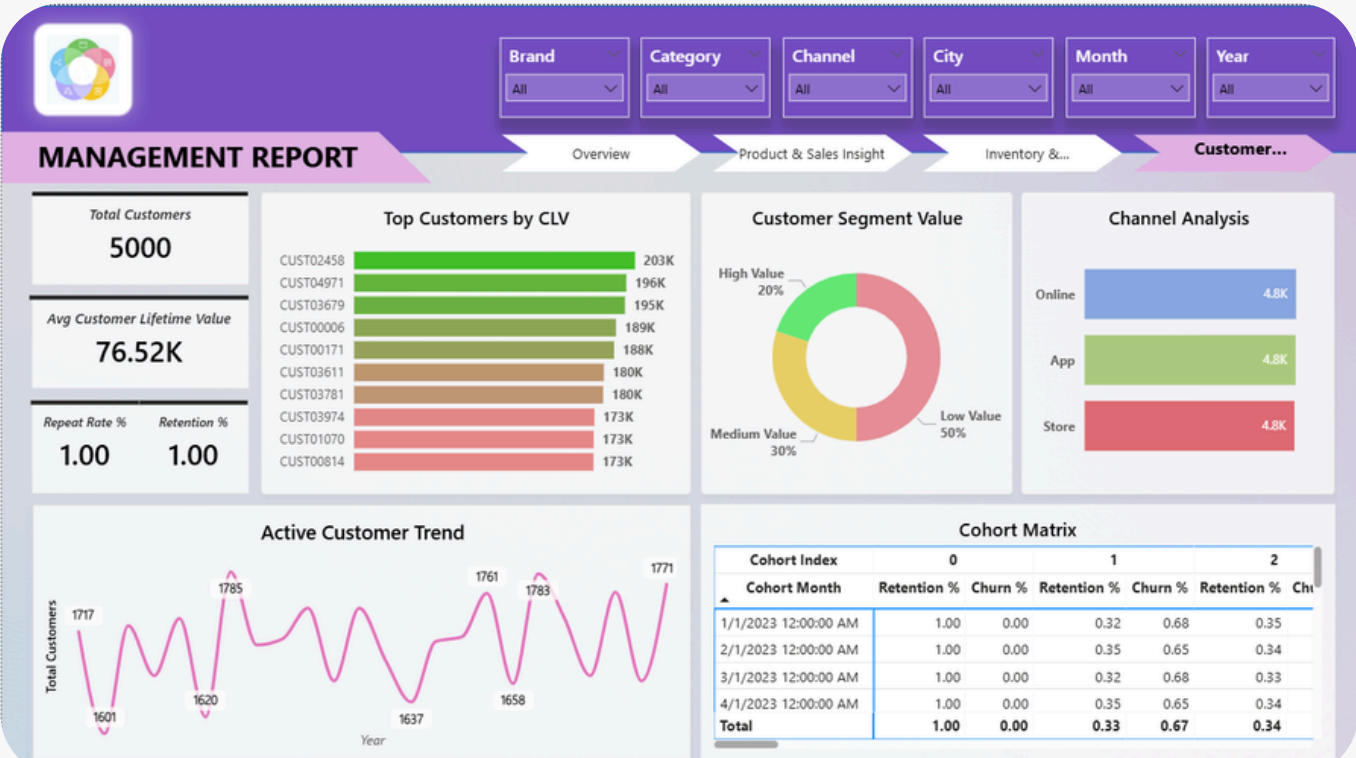
## PRODUCT AND SALES INSIGHT



## INVENTORY AND OPERATION



## CUSTOMER ANALYSIS



# KEY INSIGHT

## DISCLAIMER

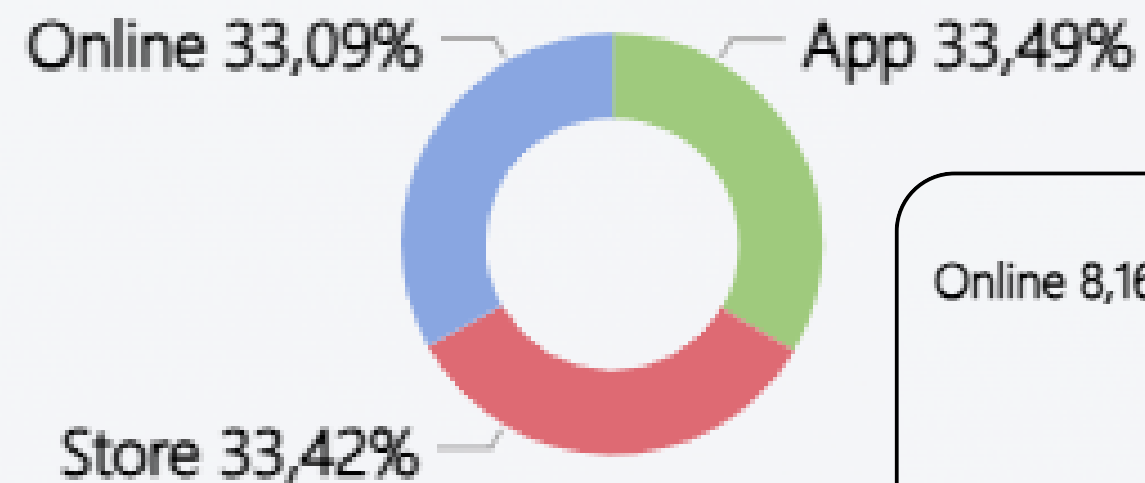
**All metrics shown in these slides are based on a synthetic AI dataset. Numerical extremes are reflective of artificial modeling rather than real-world conditions. Results are for presentation and framework review purposes only.**



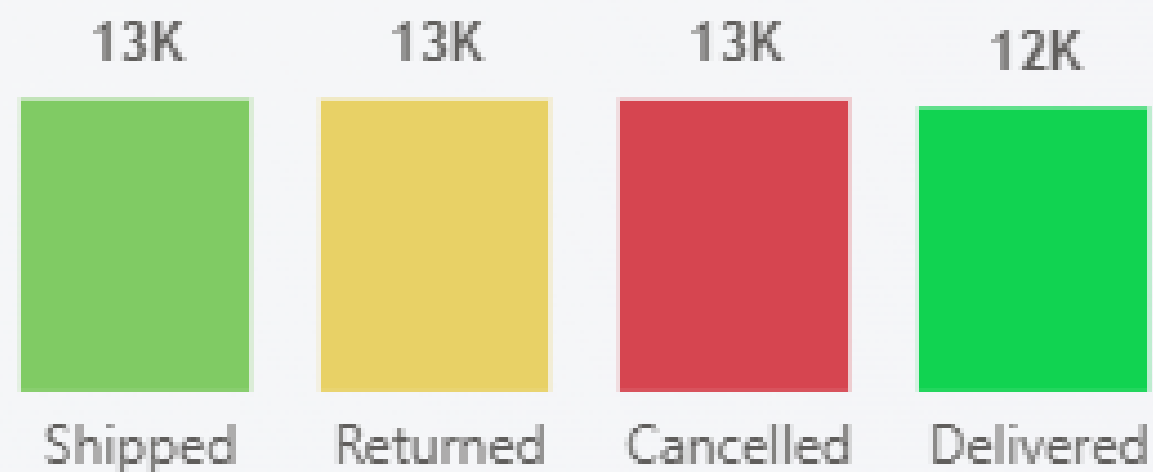
# REVENUE AND CHANNEL

## STRONG DIGITAL DEMAND MASKED BY CRITICAL ORDER FULFILLMENT ISSUES

### Revenue by Channel



### Total Orders by Status

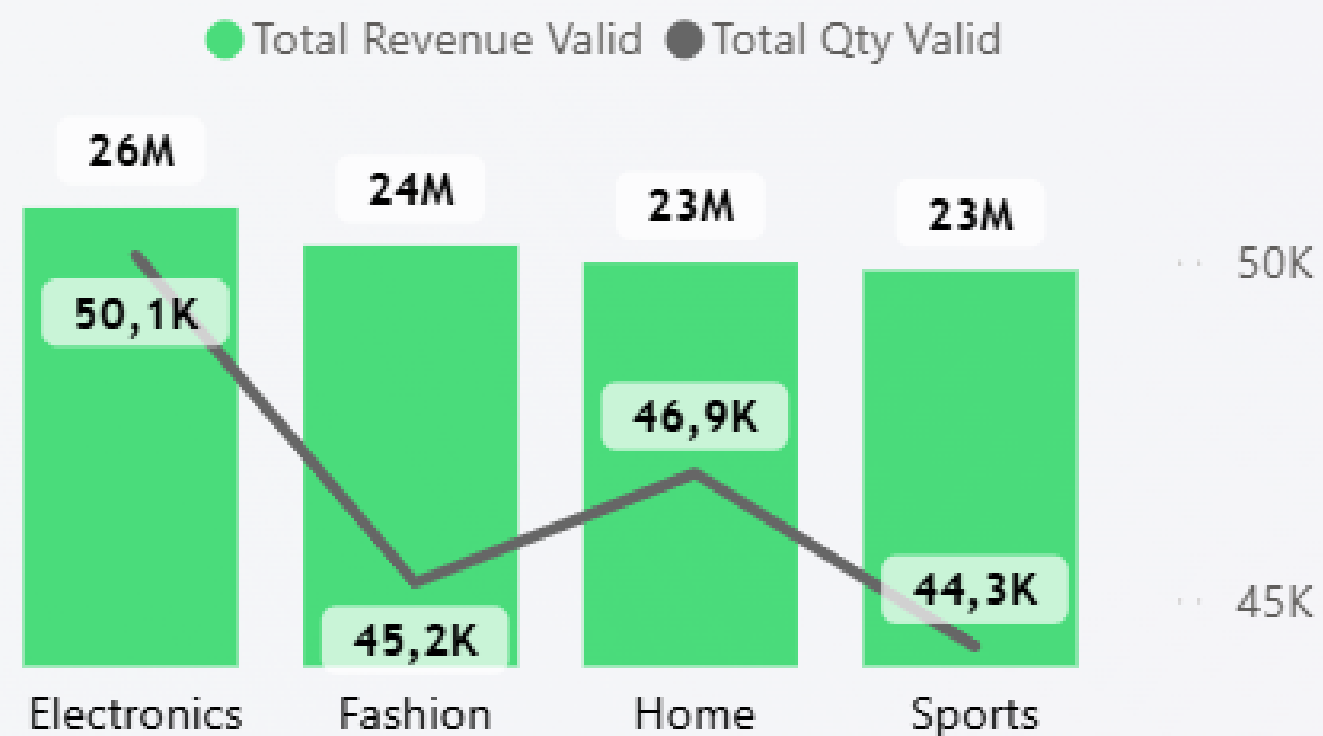


Despite **strong digital traction with Online and App channels** contributing over 66% of total revenue, the high volume of Cancelled and Returned orders signals a critical bottleneck in the fulfillment process. **The "Delivered" revenue represents only a fraction of potential earnings** up to 8.43% each of total Revenue, suggesting that while the digital storefront effectively captures interest, operational inefficiencies or product quality issues are causing significant profit leakage. Highlighting an urgent need to **improve order success rates** to ensure sustainable growth.

# REVENUE AND SALES

## SMALL VOLUME, MASSIVE IMPACT: FASHION'S HIGH-VALUE ADVANTAGE

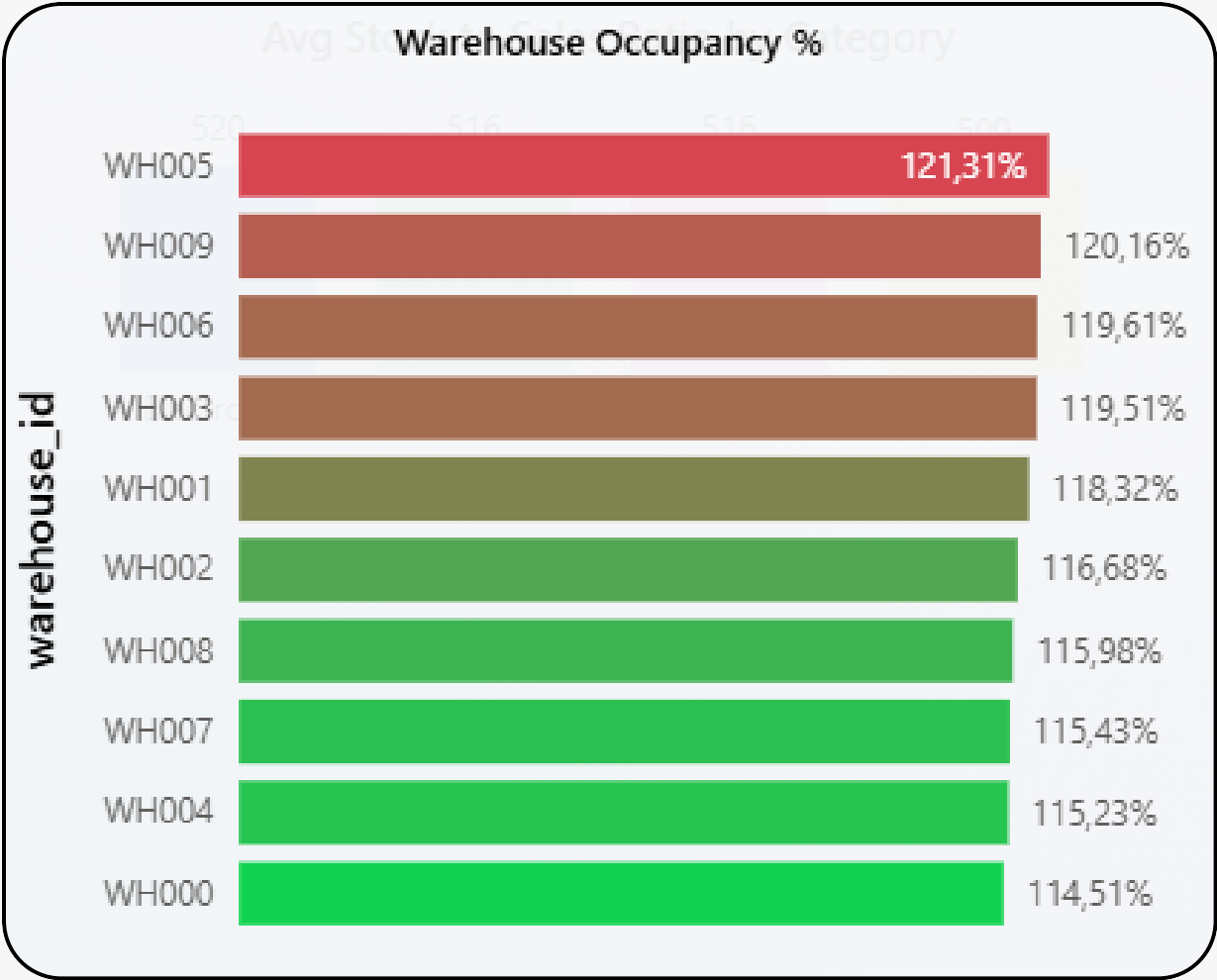
Valid Revenue and Qty Sold by Category



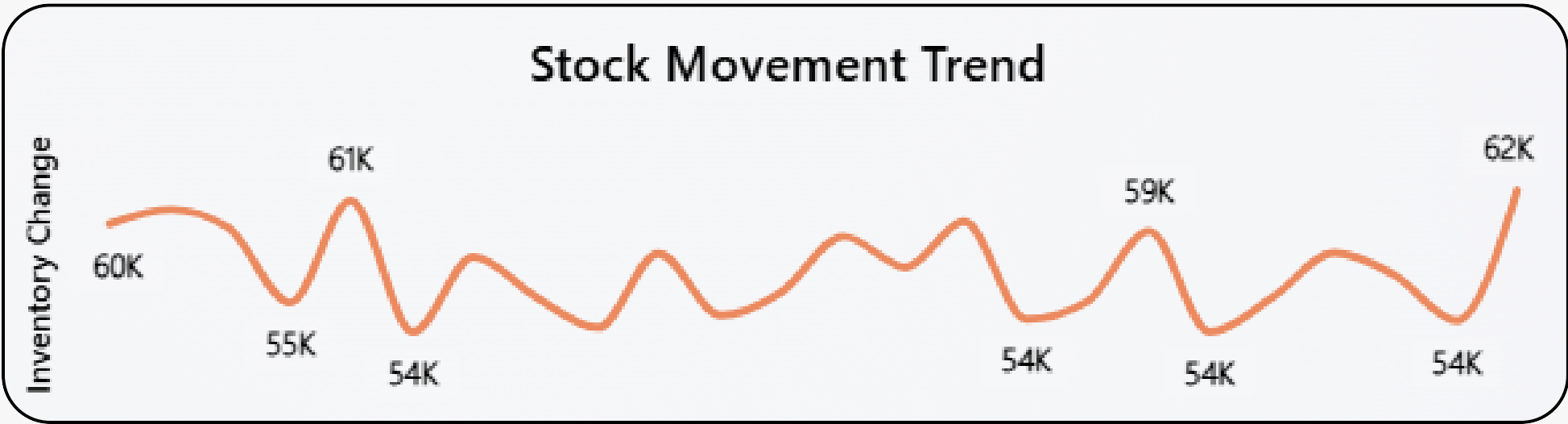
The **Fashion category** is a masterclass in efficiency, securing a **staggering 24M in revenue while moving significantly fewer units 45.2K** compared to other categories. It effectively outearns the higher-volume Home segment despite selling nearly \$2,000\$ fewer items, showcasing a superior revenue-per-unit ratio. This highlights Fashion as a **premium engine for the business**, proving that high-value transactions can drive top-tier financial performance without relying on sheer sales volume.

# INVENTORY AND OPERATION

## CRITICAL OVERCAPACITY AMID UNSTABLE INVENTORY INFLOWS



The organization is currently facing a severe storage crisis, with **every warehouse exceeding its physical capacity**, peaking at 121.31%. While the Stock Movement Trend shows active sales dips to 54K, these gains are immediately erased by **aggressive restocking spikes** that recently surged to 62K. This "yo-yo" pattern suggests that inventory is being forced into facilities that have no room to breathe. Without an **immediate pause on new intakes** or an **urgent expansion of storage space**, the supply chain remains at a breaking point, risking both operational efficiency and workplace safety.



# INVENTORY AND OPERATION

## RECLAIMING FROZEN CASH FLOW BY RESOLVING CRITICAL INVENTORY STAGNATION

515.37

**AVG Sales To  
Stock Ratio**

0.26

**AVG Inventory  
Turn Over Index**

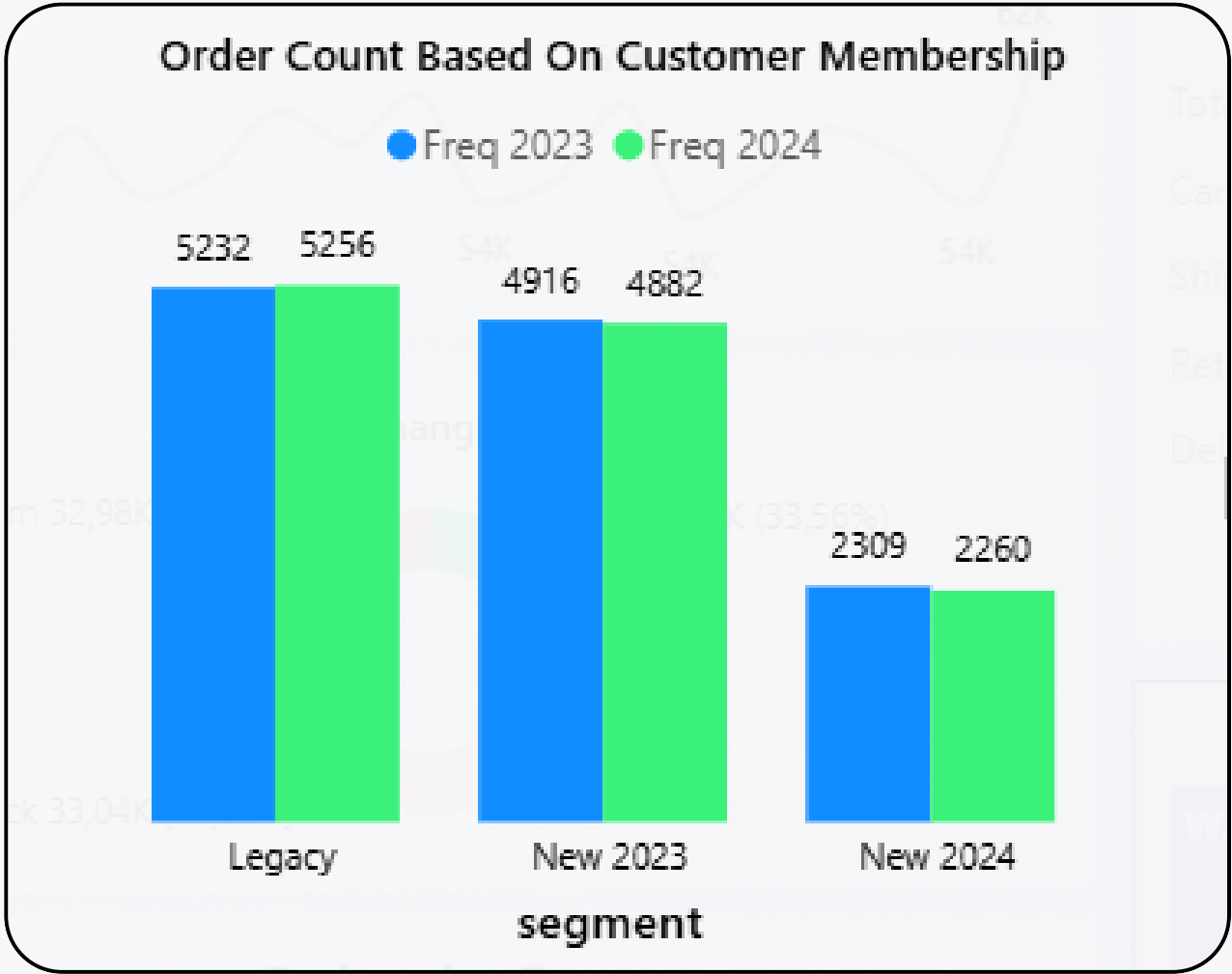
The current inventory metrics reveal a critical **imbalance between stock levels and market demand**. With an Average Stock to Sales Ratio of 515.37, the **business is carrying over 500 times the inventory required** to meet monthly sales, indicating massive overstocking. This inefficiency is further highlighted by a very low Average Turnover Index of 0.26, where nearly **half of the product catalog moves at a rate below 0.25**. This "frozen capital" increases holding costs and the risk of dead stock, as products sit in warehouses for extended periods without converting into revenue.

# CUSTOMER AND SALES

## STABLE LOYALTY AMIDST SLOWING NEW MEMBER GROWTH

|                |          |                                                 |      |
|----------------|----------|-------------------------------------------------|------|
| Total Customer | LEGACY   | Customer who sign up in 2022 (initial customer) | 2085 |
|                | NEW 2023 | Customer who sign up in 2023                    | 1975 |
|                | NEW 2024 | Customer who sign up in 2024 (new customer)     | 915  |

The company's revenue remains anchored by highly **loyal Legacy and 2023 cohorts**, who have maintained consistent transaction frequencies year-over-year. However, **a significant 53% drop in 2024** sign-ups indicates a cooling acquisition phase. While the "stickiness" of established members provides a reliable foundation, the slowing influx of new customers highlights **an urgent need to refresh marketing strategies** and diversify the acquisition funnel to ensure future growth.





THANK  
YOU

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